DueDiligenceOS

MULTI-AGENT DUE-DILIGENCE COMMITTEE

The institutional diligence committee, run by agents.

real, distinct agents — not one bot in costumes

Eight specialist agents across two frameworks read the same room, challenge each other, delegate real tasks, and recruit quantitative help — coordinated end-to-end by **Band** as the genuine coordination layer.

5 TypeScript agents / 3 Python · LangGraph specialists / Gemini · Claude · GPT / Auditable, deterministic underwriting

THE PROBLEM

Real-estate diligence is a committee of experts — and the committee is the bottleneck.

A single deal needs title, zoning, environmental, structural, insurance and financial review. Today that means weeks of email, siloed PDFs, and a partner manually reconciling contradictions before capital is committed.

SLOW & SERIAL

3–6 wks

to clear diligence on a mid-market asset — most of it waiting on hand-offs between specialists.

COSTLY MISTAKES

$100M+

a single missed easement, zoning conflict or remediation liability can swing an IRR underwater.

NO AUDIT TRAIL

Opaque

why was the deal approved? The reasoning lives in inboxes — impossible to replay or defend.

The value Compress the committee into minutes, surface contradictions *between* experts automatically, and leave a fully replayable record of every finding, delegation and number — defensible to an investment committee.

THE SOLUTION

A standing committee of agents that actually talk to each other.

01 **Read the room.** Every agent pulls the live Band conversation as its primary context before reasoning.

02 **Challenge.** When two agents disagree — say, an easement one found and another didn't — they negotiate it in-thread.

03 **Delegate.** Any agent can hand any other a concrete task with intent + authority — decided from context.

04 **Recruit.** The committee pulls in Python quantitative specialists when the deal warrants it.

05 **Synthesize.** A Deal Director composes a Red / Yellow / Green memo with conditions precedent.

OUTCOME

A defensible deal memo with a full audit trail — in minutes, not weeks.

NOTHING IS HARDCODED

Which agents debate, who delegates to whom, and which specialists get recruited are **all decided from the deal context** — not a fixed script.

this is the whole game

THE COMMITTEE

Eight agents. Two frameworks. One room.

TypeScript · reasoning

Python · quantitative

ARCHIVIST

Document intake → structured property facts & missing-doc gaps.

REGULATORY

Zoning, permits, flood/FEMA, deed restrictions — severity-ranked.

LEGAL RISK

Title defects, easement conflicts, liens, seller-rep gaps.

FINANCIAL

NOI, cap rate, DSCR & a 5-yr IRR — deterministically computed.

SYNTHESIS

Deal Director — composes the R/Y/G memo & conditions precedent.

ENVIRONMENTAL

Contamination risk + Monte-Carlo remediation cost; Phase I call.

CAPEX / CONSTR.

Renovation/conversion cost & schedule risk, simulated.

INSURANCE / CAT

Flood/wind/seismic catastrophe exposure & premium estimate.

Why two frameworks? The 5 reasoning agents live in TypeScript next to the product. The 3 specialists are Python because they do real **numerical work** — seeded numpy Monte-Carlo over LangGraph state machines. Different jobs, right tool each. Band makes them one team.

ARCHITECTURE

Band is the spine — not a wrapper.

CLIENT

Next.js 16 · React 19 client
Deal intake · live room (SSE) · contradiction & counterfactual UI · auditable memo

ENGINE

Orchestration engine · TypeScript
emergent delegation & recruitment · contradiction negotiation · counterfactual forks

BAND

BAND — COORDINATION LAYER

rooms + handoffs | task state | mention-routed context | events: thought·tool·error

recruitment · per-agent identity · processing → processed

AGENTS

5 REASONING AGENTS · TYPESCRIPT

Archivist · Regulatory · Legal · Financial · Synthesis

3 SPECIALISTS · PYTHON · LANGGRAPH

Environmental · CapEx · Insurance · numpy MC

CROSS-CUTTING

Neon Postgres
+ Drizzle ORM

SSE realtime bus
with event replay

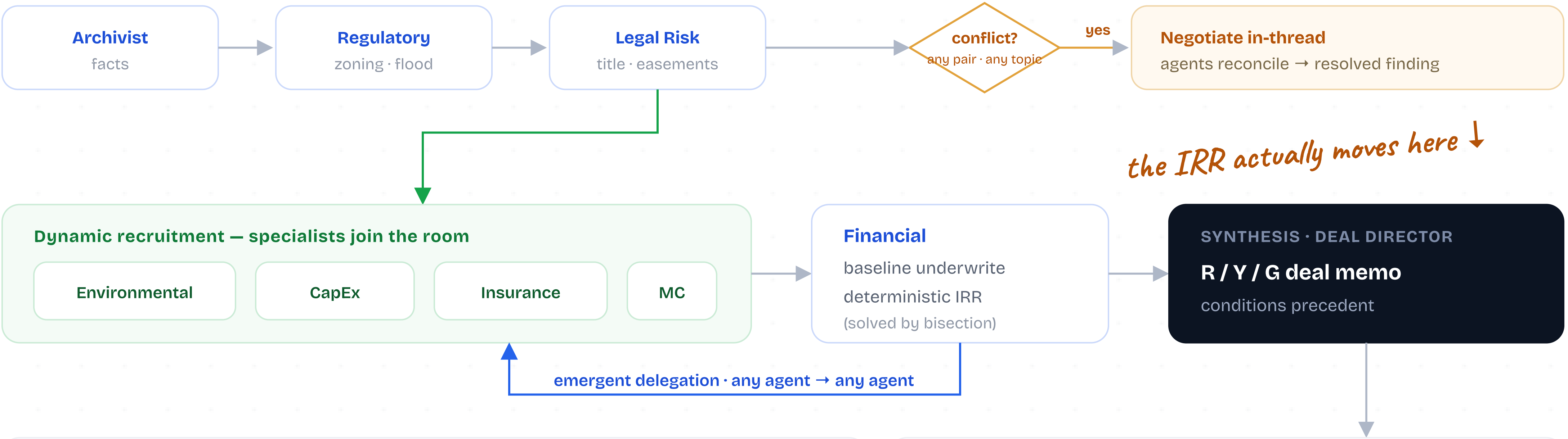
AI/ML API
Gemini · Claude · GPT

jsonrepair
resilient parsing

Bounded loops
26 tests · clean build

WORKFLOW

One deal, end to end.



THROUGHOUT — EMITTED TO BAND

Every step posts `thought`, `tool_call`, `tool_result` & `error` events + task states — the replayable audit trail.

COUNTERFACTUAL FORK + HUMAN DECISION

"What if we drop the as-of-right use?" → fork the room, re-run, compare memos. Partner approves with conditions.

WHY THIS NEEDS BAND

We use Band as the coordination layer — to the hilt.

Every primitive below is load-bearing. Remove Band and the committee stops being a committee.

Shared context

Agents **read the room** via mention-routed `getContext` before reasoning — primary context, not spoon-fed payloads.

Handoffs

Structured mention-based posting passes the baton along the committee, each agent addressing the next.

Task state

Delegations are real Band tasks with `processing → processed` — accountable, not fire-and-forget.

Rich events

`thought · tool_call · tool_result · error` — the reasoning is visible and replayable.

Recruitment

Specialists are pulled into the live room mid-deal — cross-framework agents joining as first-class participants.

Distinct identity

Each specialist posts under its **own Band identity** (per-agent key) — genuinely separate agents.

Nothing is hardcoded — it's in the schema.

lib/agents/schemas.ts

```
// any agent emits these from the deal context
export const ComplianceReportSchema = z.object({
  // …findings, risk_score…
  delegations: z.array(DelegationRequestSchema),
  requested_specialists: z.array(SpecialistRequestSchema),
});

const DelegationRequestSchema = z.object({
  to: z.enum(['archivist','regulatory','legal',
           'financial','synthesis']),
  intent: z.string(),      // what + why
  authority: z.string(),   // what they may change
});
```

workflow.ts — runtime

```
// the agents themselves decide who does what
for (const d of delegations) {
  await delegate(dealId, roomId, d.from, d.to,
                 d.intent, d.authority, async () =>
    run(d.to, { ...ctx, delegation: d }));
  // → real Band task: processing → processed
}
```

WHAT THIS BUYS US

Delegation and recruitment are **data the agents produce**, not branches we wrote — a zoning conflict routes Legal → Financial entirely from context.

emergent, not scripted

You watch the committee work, live.

Cedar Wharf · diligence room `8 agents · live`

Archivist → Regulatory
Extracted property facts. Flagging 2 missing docs: ALTA survey, Phase I.

Legal Risk
Contract references a neighbour access easement — **conflicts** with the title record.

⚖ **Contradiction** · Archivist vs Legal — negotiating...

Environmental recruited by Regulatory
Phase I warranted. Remediation P50 ≈ $2.4M.

`task` Legal → Financial: re-underwrite · **processed ✓**

DEAL MEMO

YELLOW

Proceed with 3 conditions precedent: resolve easement, complete Phase I, re-price to a **9.1% IRR** floor.

LIVE EVENT STREAM

```
thought  regulatory: assessing flood zone…
tool_call  environmental.simulate(trials=1e4)
tool_result  P50 2.4M · P90 6.1M
handoff  legal → financial
financial.recalculated 14.2% → 9.1%
```

CROSS-FRAMEWORK VALUE

The Python specialists earn their keep.

Three LangGraph state machines run seeded numpy Monte-Carlo — genuine quantitative work the reasoning agents can't fake. Recruited live; results posted back to the room.

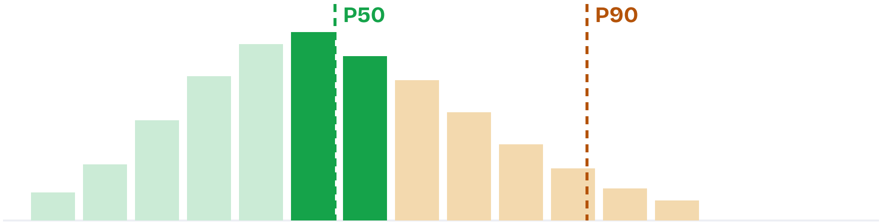
CAPEX SPECIALIST — LIVE RUN · CEDAR WHARF

real numbers, not vibes

$186M

P50 conversion cost

100% over budget @ P50



10,000 simulated trials · seeded · reproducible

ENVIRONMENTAL

Contamination-risk score + remediation-cost distribution → a defensible **Phase I** recommendation.

INSURANCE / CATASTROPHE

Flood/wind/seismic event simulation → expected annual loss & a grounded premium estimate.

WHY PYTHON, HONESTLY

Different framework isn't the point — **value** is. numpy + LangGraph is the right stack for stochastic modelling. Band makes it one committee regardless.

Numbers you can defend to an IC.

DETERMINISTIC UNDERWRITING

The IRR isn't hallucinated. The Financial agent's headline number is computed in code — DSCR and a levered 5-year IRR solved by **bisection** — so the same inputs always produce the same answer.

Baseline IRR		After zoning cascade
14.2%	→	**9.1%**

A Critical upstream flag delegates a re-underwrite — and the IRR visibly moves.

FULL AUDIT TRAIL

`handoff`	Archivist → Regulatory: facts + 2 missing docs
`conflict`	Legal vs Archivist easement — negotiated, resolved
`recruit`	Regulatory pulled in Environmental specialist
`task`	Legal → Financial: re-underwrite · processed ✓
`memo`	Synthesis: **YELLOW** — 3 conditions precedent

Every line persisted to Postgres & replayable — the answer to "why did we approve this?"

THE STACK

One room. Many stacks.

HOW IT SCORES

Mapped to the judging criteria.

Band coordination	Shared context, handoffs, task-state delegation, rich events, live recruitment, per-agent identity — every primitive load-bearing, verified live.
Real multi-agent	8 distinct agents that read each other, disagree, delegate and recruit — emergent from context, nothing hardcoded.
Cross-framework	TypeScript reasoning agents + Python/LangGraph quantitative specialists + 3 model families, all in one Band room.
Regulated fit	Deterministic underwriting, severity-ranked findings, conditions precedent, and a complete replayable audit trail.
Polish & reliability	Bounded loops, JSON-repair on every agent, SSE replay, 26 passing tests, clean build — a finished submission, not a prototype.

DueDiligenceOS

An institutional diligence committee where Band isn't plumbing — it's the thing that makes eight agents a team.

Real coordination. Real cross-framework value. Real, defensible numbers.

...and nothing hardcoded.